finaldelim

Delimiter sizing, nesting colors, and equation breaks

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finaldelim adds three features to LuaLaTeX mathematics: distance-weighted delimiter sizing, nesting colors named `niji`, and explicit line breaks directly inside `equation`.

Compile with **LuaLaTeX** and load the package:

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math}
\usepackage{finaldelim}
```

The new sizing policy is enabled by default. The manual uses Latin Modern Math; finaldelim also works with other math fonts. Optional continuous-outline rendering is described in Section 1.3. It loads `amsmath` and `xcolor` automatically.

1 Delimiter sizing

Native `\left` and `\right` size a pair from its tallest enclosed content. A distant fraction can therefore make both delimiters large beside much smaller neighbours. finaldelim weights content by its distance from each edge, balancing the pair with the surrounding expression.

The correction factor in Stirling's series mixes low terms with taller fractions.[1] Here $C(n) = n!/[\sqrt{2\pi n}(n/e)^n]$.

Native sizing

LaTeX source

```
\begin{equation*}[sizing=native]
 C(n)\sim\left(1+\frac{1}{12n}
 +\frac{1}{288n^2}
 -\frac{139}{51840n^3}
 +\cdots\right).
\end{equation*}
```

Result

$$C(n) \sim \left(1 + \frac{1}{12n} + \frac{1}{288n^2} - \frac{139}{51840n^3} + \cdots\right).$$

Default weighted sizing

LaTeX source

```
\begin{equation*}
 C(n)\sim\left(1+\frac{1}{12n}
 +\frac{1}{288n^2}
 -\frac{139}{51840n^3}
 +\cdots\right).
\end{equation*}
```

Result

$$C(n) \sim \left(1 + \frac{1}{12n} + \frac{1}{288n^2} - \frac{139}{51840n^3} + \cdots\right).$$

[1] Wikipedia, Stirling's approximation, Stirling series.

1.1 Why distance, width, and height matter

The model treats the contents as a horizontal profile. A measured box of height h_i and depth d_i has radius

$$r_i = \max(h_i - a,\, d_i + a),$$

where a is the math axis height. Its radius applies over its actual width. This gives a broad tall term more influence than a narrow one at the same height. Spaces use the smallest measured radius in the group.

Let $r(x)$ be this profile over the content width W. The two delimiters see opposite distance weights:

$$w_L(x) = e^{-x/\lambda}, \qquad w_R(x) = e^{-(W-x)/\lambda}.$$

At distance λ, a contribution has $1/e$ of its weight at the edge. The left and right radius estimates are

$$R_L = \left(\frac{\int_0^W r(x)^q w_L(x)\,dx}{\int_0^W w_L(x)\,dx} \right)^{1/q}, \qquad R_R = \left(\frac{\int_0^W r(x)^q w_R(x)\,dx}{\int_0^W w_R(x)\,dx} \right)^{1/q}.$$

The larger estimate sets a shared target radius:

$$R = s \max(R_L, R_R), \qquad s = \texttt{scale}.$$

The power q gives tall content extra influence even when it occupies only a small part of the width. Taking the maximum prevents a distant delimiter from diluting the estimate at a delimiter close to tall content. TeX then selects actual delimiters from the math font using this radius.

1.2 Choosing settings

Option	Default	Effect
`lambda`	`20pt`	A shorter decay length makes nearby content more decisive.
`q`	`8`	A larger value gives tall content more influence.
`scale`	`0.9`	Multiplies the target radius about the math axis.

The default decay length is a physical distance: 20pt at every font size. `lambda` and `scale` must be positive; `q` must be nonnegative. The special value `q=0` selects the maximum-radius limit.

```
\usepackage[lambda=20pt,q=8,scale=0.9]{finaldelim}
% Change settings for subsequent formulas:
\FinalDelimSetup{scale=1} % use the full estimated radius
% Return to native sizing; colors and line breaks remain available:
\FinalDelimSetup{sizing=native}
\FinalDelimSetup{sizing=weighted,lambda=20pt}
```

For a local setting, write:

```
\begin{equation}[lambda=30pt]
```

The default scale of 0.9 favours a compact visual frame rather than requiring the delimiters to enclose the full height of the contents. The scale applies to weighted and endpoint sizing. It changes the sizing target, not the glyph outline; font size steps can leave a glyph unchanged. Native sizing and explicitly sized delimiters are unaffected.

1.3 Continuous delimiter outlines

The sizing model chooses how large a delimiter should be. Continuous-outline rendering gives it intermediate sizes between the font's prebuilt glyphs and fills assembled delimiters as a single outline, avoiding repeated painting at component joins. It uses the current math font's own shapes.

Enable this optional experimental feature after loading finaldelim:

```
\usepackage{finaldelim}
\usepackage{experimental/finaldelim-draw}
```

Keep `finaldelim-draw.sty`, `finaldelim-draw.lua` and `finaldelim-outline-fonts.lua` in an `experimental/` subdirectory beside the document. The renderer loads font outlines automatically.

Then write ordinary `\left`, `\middle` and `\right` commands. Niji and line breaks work with the drawn outlines too. Superscripts and subscripts after a visible closing delimiter follow its final size; the contents retain their full height and depth for surrounding layout.

This four-by-two matrix illustrates the Frobenius norm.[2] Both results below use the same default sizing settings.

Native font glyphs

LaTeX source

```
\begin{equation*}
 \left\Vert\begin{matrix}
 a_{11}&a_{12}\\ a_{21}&a_{22}\\
 a_{31}&a_{32}\\ a_{41}&a_{42}
 \end{matrix}\right\Vert_F
 =\sqrt{\sum_{i=1}^4\sum_{j=1}^2
              |a_{ij}|^2}
\end{equation*}
```

Result

$$\left\Vert\begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \end{matrix}\right\Vert_F = \sqrt{\sum_{i=1}^4\sum_{j=1}^2 |a_{ij}|^2}$$

Continuous outlines

LaTeX source

```
\begin{equation*}
 \left\Vert\begin{matrix}
 a_{11}&a_{12}\\ a_{21}&a_{22}\\
 a_{31}&a_{32}\\ a_{41}&a_{42}
 \end{matrix}\right\Vert_F
 =\sqrt{\sum_{i=1}^4\sum_{j=1}^2
              |a_{ij}|^2}
\end{equation*}
```

Result

$$\left\Vert\begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \end{matrix}\right\Vert_F = \sqrt{\sum_{i=1}^4\sum_{j=1}^2 |a_{ij}|^2}$$

`\FinalDelimDrawOff` returns to native glyph rendering; `\FinalDelimDrawOn` enables drawing again. These commands respect TeX groups and do not change the sizing settings.

Between prebuilt sizes, the renderer interpolates the font's contours. For larger assembled symbols it keeps the font's component placement but paints one combined path. Slashes, backslashes, single angles and double angles can extend beyond the largest prebuilt glyph: their width follows the recent size trend, while stroke thickness and terminal details are preserved. Double angles retain their inter-stroke spacing. Other symbols without larger variants or assembly parts retain their native size limit. Transformed fonts and explicit variable-font instances currently retain native rendering. Drawn delimiters are PDF paths; text extraction and accessibility tagging for these paths are not yet supported.

[2] Wikipedia, Matrix norm, Frobenius norm; shown here for a four-by-two matrix.

1.4 One or more middle delimiters

Use `\middle` for a delimiter inside a `\left/\right` group. Every middle delimiter participates in the size calculation. A matrix element in bra–ket notation uses two of them.[3]

LaTeX source

```
\begin{equation*}[niji]
 \left\langle\phi
 \middle|\hat A\middle|\psi
 \right\rangle
\end{equation*}
```

Result

$$\left\langle\phi\middle|\hat A\middle|\psi\right\rangle$$

The outer delimiters and both bars share one sizing target. The math font supplies the actual glyphs. With `niji`, the middle delimiters receive the same color as their surrounding pair.

Each delimiter sees its own neighbourhood

Let s_j be the horizontal position of delimiter j, including both ends and every middle delimiter. Its weight and radius estimate are

$$w_j(x) = e^{-|x-s_j|/\lambda}, \qquad R_j = \left(\frac{\int_0^W r(x)^q w_j(x)\,dx}{\int_0^W w_j(x)\,dx} \right)^{1/q}.$$

For N delimiters, their shared radius is

$$R = s \max_{1\le j\le N} R_j.$$

A middle delimiter receives contributions from both sides. Moving it closer to tall content can increase its estimate and therefore the shared size. With only the two outer delimiters, this is the earlier left/right model.

Across line breaks

Middle delimiters stay at their source positions when the formula is split into rows. Under `spanning=endpoints`, the size envelope includes every row carrying one of the group's delimiters: the first row, the last row, and any row containing a middle delimiter. Under `spanning=full`, all delimiter positions use the distance-weighted model over the continuous contents. Nested groups are sized independently.

[3] Wikipedia, Bra–ket notation, linear operators acting on bras.

2 Niji: colors for nested delimiters

Enable `niji` to give matching delimiters the same nesting color:

```
\usepackage[niji]{finaldelim}
```

`rainbow` is an alias. Colors apply to explicit `\left`/`\right` pairs; a `\middle` delimiter follows its pair. Ordinary parentheses and the mathematical contents keep their usual color.

The normal cumulative distribution function has a bracketed expression with an inner argument of the error function.[4]

Without niji

LaTeX source

```
\begin{equation*}[niji=false]
 F(x)=\frac12\left[
 1+\operatorname{erf}\left(
 \frac{x-\mu}{\sigma\sqrt2}
 \right)\right].
\end{equation*}
```

Result

$$F(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right].$$

With niji

LaTeX source

```
\begin{equation*}[niji]
 F(x)=\frac12\left[
 1+\operatorname{erf}\left(
 \frac{x-\mu}{\sigma\sqrt2}
 \right)\right].
\end{equation*}
```

Result

$$F(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right].$$

2.1 Inline mathematics

The same setting works in ordinary inline math.

LaTeX source

```
\FinalDelimSetup{niji}
The standardized argument is
$\left(\frac{x-\mu}{\sigma}
\right)$.
```

Result

The standardized argument is $\left(\frac{x-\mu}{\sigma}\right)$.

2.2 Settings and palettes

```
\FinalDelimSetup{niji=false}            % subsequent mathematics
\FinalDelimSetup{niji,colors={blue,red,teal}}
```

Use an environment option, such as `\begin{equation}[niji=false]`, to override the setting for one display. The palette repeats at deeper nesting levels. Sizing and coloring can be selected independently.

[4] Wikipedia, Normal distribution, cumulative distribution function.

3 Line breaks directly in equation

Place `\\` at a break and `&` at the alignment point. Keep the original delimiter pair: `\left` can open on one row and `\right` can close on another.

3.1 The same long formula, old and new

The divergence between two multivariate normal distributions contains a trace, a quadratic form, and a logarithmic determinant.[5] Keeping the full notation makes this a useful case for several rows: the left-hand side introduces the quantity, and each subsequent row carries a term.

Plain amsmath: split and invisible delimiters

The traditional form uses an inner `split`. The opening bracket is paired with `\right.`; the closing bracket is paired with `\left.`. The intermediate row needs no delimiter commands. A `\vphantom` copies the height and depth of the final logarithmic fraction into the opening row, so both brackets receive the same size. **Red bold** marks the extra code.

LaTeX source

```
\begin{equation}
 \begin{split}
 &D_{\mathrm{KL}}(\mathcal N_0
     \Vert\mathcal N_1)\\
 &=\frac12\left[
   \vphantom{\ln\frac{\det\Sigma_1}
                     {\det\Sigma_0}}
   \operatorname{tr}(
     \Sigma_1^{-1}\Sigma_0)-k\right.\\
 &\quad+(\mu_1-\mu_0)^{\mathsf T}
   \Sigma_1^{-1}(\mu_1-\mu_0)\\
 &\quad\left.+\ln\frac{\det\Sigma_1}
                  {\det\Sigma_0}
   \right].
 \end{split}
\end{equation}
```

Result

$$
\begin{aligned}
&D_{\mathrm{KL}}(\mathcal{N}_0 \| \mathcal{N}_1) \\
&= \frac{1}{2}\left[\operatorname{tr}(\Sigma_1^{-1}\Sigma_0) - k \right. \\
&\quad + (\mu_1 - \mu_0)^{\mathsf{T}} \Sigma_1^{-1} (\mu_1 - \mu_0) \\
&\quad \left. + \ln \frac{\det \Sigma_1}{\det \Sigma_0} \right].
\end{aligned}
\tag{1}
$$

finaldelim: direct breaks, with niji included

Keep the same break points, remove the extra wrapper, invisible delimiters, and height phantom, and keep a single `\left`/`\right` pair across the rows.

LaTeX source

```
\begin{equation}[niji]
 &D_{\mathrm{KL}}(\mathcal N_0
     \Vert\mathcal N_1)\\
 &=\frac12\left[
   \operatorname{tr}\left(
     \Sigma_1^{-1}\Sigma_0\right)-k\\
 &\quad+(\mu_1-\mu_0)^{\mathsf T}
   \Sigma_1^{-1}(\mu_1-\mu_0)\\
 &\quad+\ln\frac{\det\Sigma_1}
                  {\det\Sigma_0}
   \right].
\end{equation}
```

Result

$$
\begin{aligned}
&D_{\mathrm{KL}}(\mathcal{N}_0 \| \mathcal{N}_1) \\
&= \frac{1}{2}\left[\operatorname{tr}\left(\Sigma_1^{-1}\Sigma_0\right) - k \right. \\
&\quad + (\mu_1 - \mu_0)^{\mathsf{T}} \Sigma_1^{-1} (\mu_1 - \mu_0) \\
&\quad \left. + \ln \frac{\det \Sigma_1}{\det \Sigma_0} \right].
\end{aligned}
\tag{2}
$$

Niji is included as an optional companion feature: adding `[niji]` colors matching delimiters by nesting level, including pairs that cross rows. The example also uses `\left`/`\right` around the trace argument so its inner pair receives the next color. Both forms keep one equation number.

[5] Wikipedia, Kullback–Leibler divergence, multivariate normal distributions.

3.2 Break a long expansion into readable groups

The cubic trinomial expansion has ten terms.[6] Breaking before groups of mixed terms makes it easier to scan while preserving one equation number. The source contains only `equation`, alignment markers, and the chosen breaks.

LaTeX source

```
\begin{equation}
 (a+b+c)^3
 &=a^3+b^3+c^3\\
 &\quad+3a^2b+3a^2c+3b^2a\\
 &\quad+3b^2c+3c^2a+3c^2b\\
 &\quad+6abc.
\end{equation}
```

Result

$$\begin{aligned}(a+b+c)^3 &= a^3+b^3+c^3\\ &\quad+3a^2b+3a^2c+3b^2a\\ &\quad+3b^2c+3c^2a+3c^2b\\ &\quad+6abc.\end{aligned} \tag{3}$$

3.3 Break at a change in the calculation

The normal-distribution log-likelihood starts with a sum and then expands into logarithmic terms and a sum of squared residuals.[7] A new row separates the expansion step; another keeps the residual sum together.

LaTeX source

```
\begin{equation}
 \ln\mathcal L(\mu,\sigma^2)
 &=\sum_{i=1}^n
   \ln f(x_i\mid\mu,\sigma^2)\\
 &=-\frac n2\ln(2\pi)
   -\frac n2\ln\sigma^2\\
 &\quad-\frac1{2\sigma^2}
   \sum_{i=1}^n(x_i-\mu)^2.
\end{equation}
```

Result

$$\begin{aligned}\ln\mathcal{L}(\mu,\sigma^2) &= \sum_{i=1}^{n}\ln f(x_i\mid\mu,\sigma^2)\\ &= -\frac{n}{2}\ln(2\pi)-\frac{n}{2}\ln\sigma^2\\ &\quad-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2.\end{aligned} \tag{4}$$

3.4 Choose the spanning size

`spanning=endpoints`, the default, uses the maximum height and depth of the first and last rows inside a spanning pair, plus any rows carrying its `\middle` delimiters. `spanning=full` instead applies the weighted model to its entire continuous contents, including the middle rows. Select either policy on the display:

```
\begin{equation}[spanning=full,niji]
 ...
\end{equation}
```

For pairs contained on one row, weighted sizing applies in either mode. `\\[3pt]` adds space after a row; `\jot` controls the usual extra row spacing. Additional `&` markers create additional alignment pairs.

[6] Wikipedia, Multinomial theorem, cubic trinomial example.
[7] Wikipedia, Normal distribution, estimation of parameters.

3.5 A number for each row

Change `equation` to `align` when rows need separate numbers. The maximum-likelihood estimates of a normal distribution provide a short example.[8]

LaTeX source

```
\begin{align}
 \hat\mu
 &=\frac1n\sum_{i=1}^n x_i
   \label{eq:mean}\\
 \hat\sigma^2
 &=\frac1n\sum_{i=1}^n
   (x_i-\hat\mu)^2
   \label{eq:variance}
\end{align}
```

Result

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{5}$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \hat{\mu})^2 \tag{6}$$

The labels refer separately to the mean, (5), and the variance, (6). Use `\notag` on a row to suppress its number. `align*` suppresses all automatic numbers.

3.6 Environment guide

Environment	Use
`equation`	Direct breaks and alignment, one number for the whole formula.
`equation*`, `\[...\]`	The same layout without an automatic number.
`align`, `align*`	With or without per-row numbering.
`alignat`, `alignat*`	An explicit column-pair count, with user-controlled spacing.
`matchedaligned`	An embedded alignment, with optional `[t]`, `[c]`, or `[b]` positioning.

The explicit names `matchedsplit`, `matchedalign` and `matchedalignat` are also available. Multiple alignment pairs are supported; the author determines their mathematical interpretation.

3.7 Layout controls

finaldelim lays out the mathematics continuously and then separates the actual contents at the row and alignment markers. Fractions, scripts and embedded matrices stay together. You choose where to break.

Use `debug=boxes`, `debug=alignment`, or `debug=all` as a package or display option to inspect boxes and alignment points. Box outlines use `lua-visual-debug` and its `\lvdset` interface.

Installation

Install these five files together beside your document or in a local TEXMF tree: `finaldelim.sty`, `finaldelim-luatex.sty`, `finaldelim-interface.sty`, `finaldelim.lua`, and `finaldelim-size.lua`. This implementation targets LuaLaTeX; tagged mathematical structure for the physical line-breaking backend is not yet supported.

Maintained by Yu Zhai. LPPL 1.3c or later. Formula sources are given in footnotes; notation and line breaks are adapted for these examples. Sources consulted 4 November 2026.

[8] Wikipedia, Normal distribution, estimation of parameters.